

UNIVERSITY OF LEICESTER

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COURSE **:** ADVANCED ELECTRICAL AND ELECTRONICS ENGINEERING WITH MANAGEMENT MSC

MODULE **:** EG4217/EG7217 – ADVANCED COMMUNICATIONS

Coursework 1

Task-1

We need the values of the distance and frequency to calculate the free space loss.

The distance varies between 1 km and 60 kilometres (distance in kilometers).

Frequency in 100Mhz:

72.352 88.261 98.417 100.135 102.478 106.974 110.598

Frequency in 1000mhz:

92.43 101.682 107.829 110.462 115.763 120.367 125.952

Plot: Distance vs Loss(db)

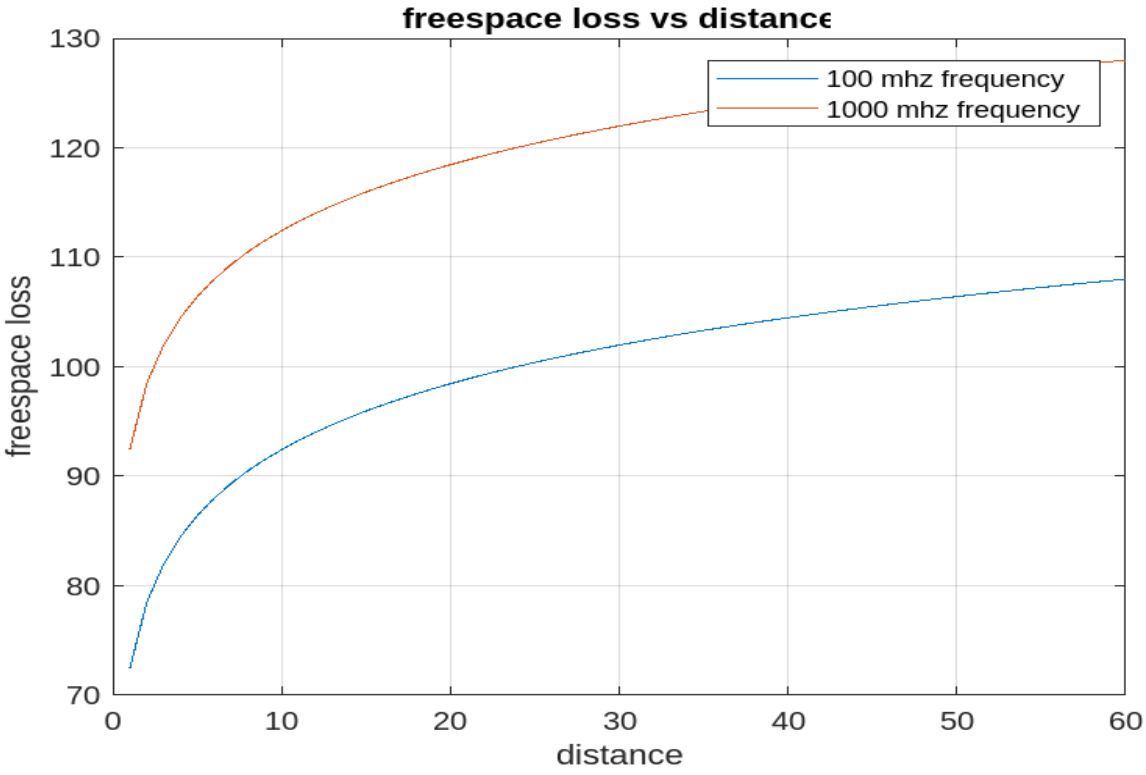


Figure 1

The signal's intensity decreases as it gets further from the transmitter, as is shown in the preceding image, which indicates that free space loss grows as distance does. The inverse square rule governs how far radio waves may travel, and this dispersion is proportional to the square of the wavelength.

k/d2 is the signal level (d- distance)

Freespace loss = (4 * pi * distance / wavelength)

Task-2

Plot: Distance vs Loss(db)

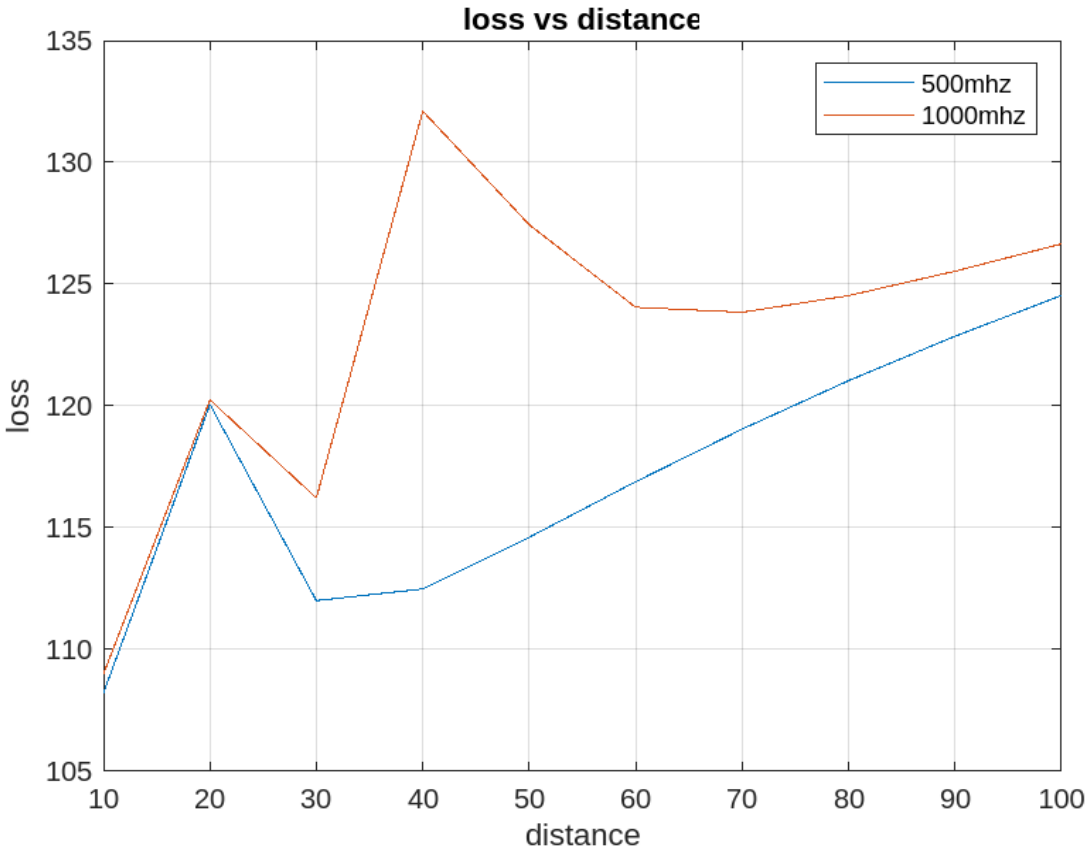


Figure 2

The loss will be smaller in comparison to the free space loss since the two-ray ground model is more sophisticated. It is also known as a multipath model since the signal travels along two routes after leaving the transmitter.

Limiting range value due to physical factors:

1)Distance:

The distance exceeds the heights of the transmitter and receiver by a wide margin.

2) Phase difference:

The received power is lower because of the significantly greater phase difference between the two photons. Due to the greater phase difference, destructive interference results, wherein the wave's positive and negative displacements cancel one another out.

Why specific distances produce peaks in loss values:

Due to the two beams' destructive combination at a specific moment, there are large peaks in loss that occur. As a result, the received power is zero and the transmitted power over received power is infinite.

Task-3

Plot: Distance vs Height of antenna(mtrs)

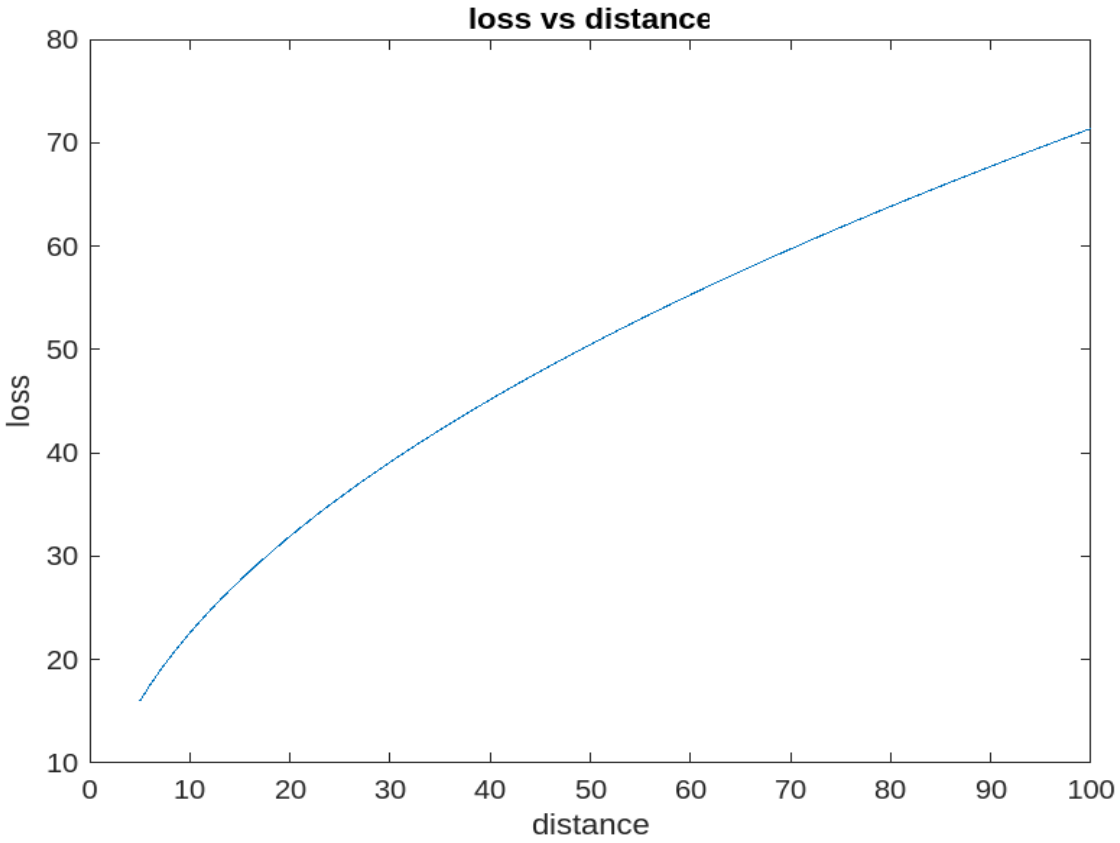


Figure 3

The antenna height vs. distance curve is seen in the above graphic.

Here, The greatest distance for line-of-site communication is

d= sqrt(2*R*ht)+sqrt(2*R*hr)

ht- height of the transmitter

hr-height of the receiver

Given that distance is directly proportional to height in the formula above, it can be shown that as height grows, so does distance.

Task-4

Diffraction Loss

We may determine the diffraction loss loss = p526 by using this function (range,ht,hr,freq,pol,surface)

Range- 100 kilometres between the transmitter and the receiver Frequency: 6 GHz (6000 MHz)

Ht = 100 m for the transmitter, Hr = 100 m for the receiver, and Pol = vertical polarisation

Ocean surface water

Plot-Diffraction vs Range

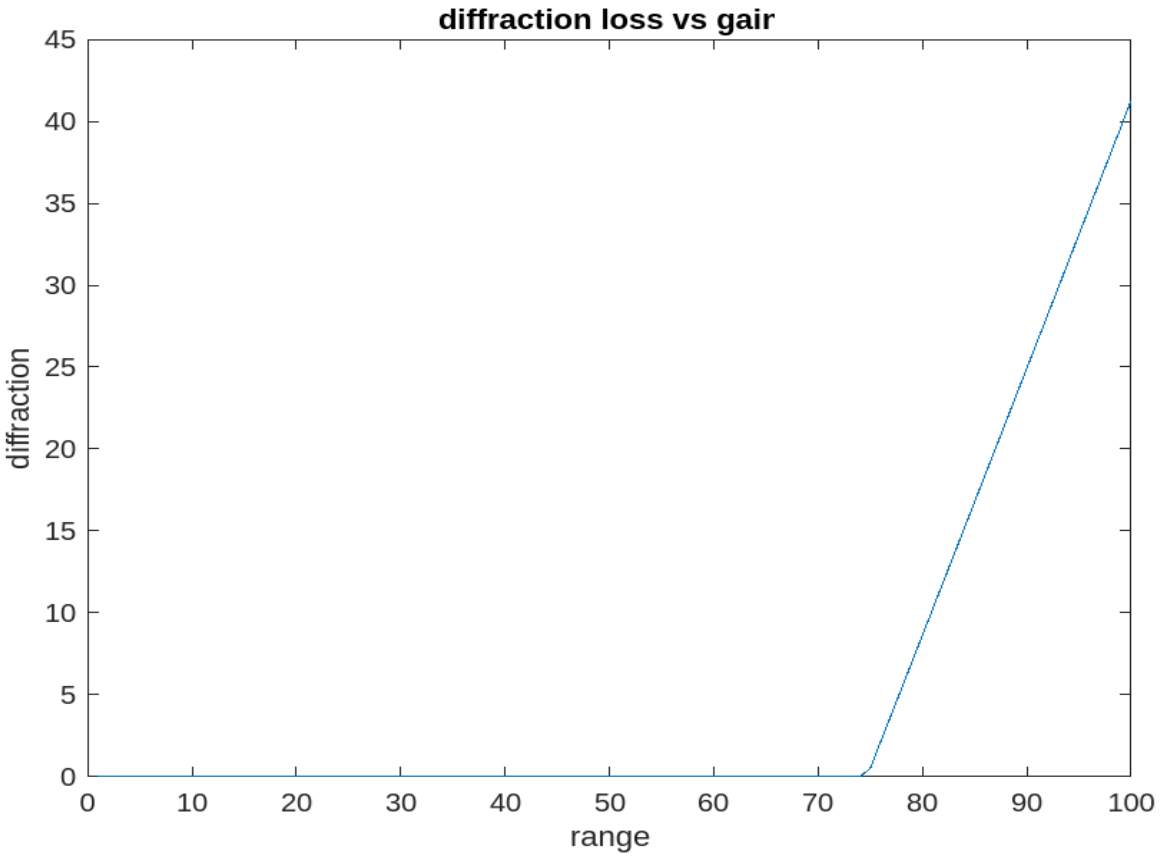


Figure 4

After increasing continuously from zero to the maximum LOS distance = 75.38 Km till the maximum distance diffraction loss is zero.

Task-5

Receiver Location	**Distance from Jersey(km)**	**Receive Antenna height(m)**	**L1**	**L2**	**L3**	**Received power dBW(Best Estimate)**
Guernsey	33	10	116.250	117.615	0.6891	-215.582
Alderney	50	9	121.895	125	16.853	-242.687
Location J2	0.3	1	74.568	70.158	0	-126.984
Portland	140.842	10	130	141.659	73.982	-374.961
Leicester	375.981	30	135.841	157.682	382.743	-714.652

Transmitted power = 30 watts

Gain of Transmitter = 2db

Gain of Receiver = 2db

Received power(dBW) = transmitted power (dbw) + tx gain + rx gain-losses

Coursework 2

Plot for Input Parameters

Distance

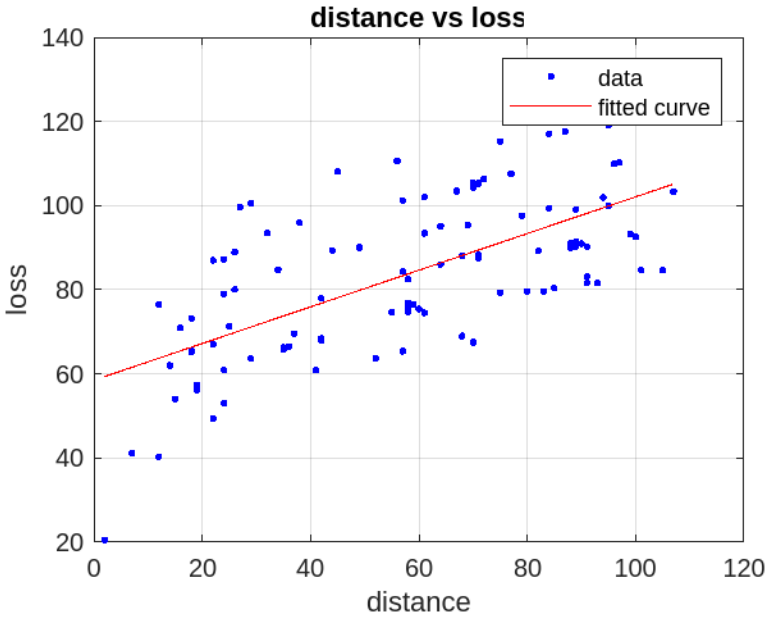


Figure 5

The best fit line is shown in red and the points scatter has a distinct trend that is roughly linear. All of the points have fallen close to this line.

The MATLAB function "fit" finds the best fit line.

```
fit1=fit(dist,loss,'poly1')
```

We advise the fit function that we wish to fit it onto a first-order linear fit.We are employing "poly1" as we require a straight line. Following the MATLAB computation, we obtain f.P1, which represents the gradient, and f.P2, which represents the intercept. The dispersion that surrounds the particular line alerts us to the fact that additional factors will affect the loss.

For fit1, gradient and intercept

```
fit1 =

     Linear model Poly1:
     fit1(x) = p1*x + p2
     Coefficients (with 95% confidence bounds):
       p1 =      0.4358  (0.3364, 0.5352)
       p2 =        58.5  (52.16, 64.84)
```

Plotting using fit1, distance, and the specified loss, the result shown above, allows us to assess the fit's quality.

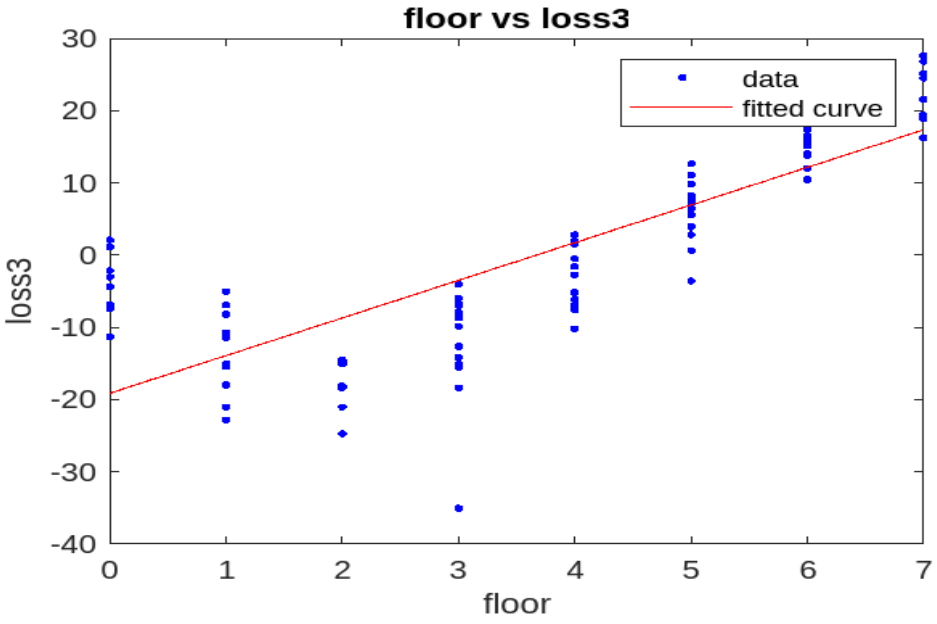
plot(fit1,dist,loss)

With the creation of a new variable loss2,

loss2=loss-fit1.p1*dist-fit1.p2

Floor

According to the floor attenuation factor (FAF), when the number of floors decreases, the FAF also gradually drops in decibels, however after six floor separations, there isn't much more route loss visible. Planning a building's construction while taking the FAF into consideration always provides a cushion.



The floor is the last input parameter, and MATLAB's fit test was used to determine the line that fits the parameter the best.

```
fit2=fit(nw,loss2,'poly1')
```

The value of p1, p2, and p3 are obtained in this case by using "poly2" to create a parabolic curve. After the MATLAB computations, the gradient and intercept are produced.

```
fit2 =

     Linear model Poly1:
     fit2(x) = p1*x + p2
     Coefficients (with 95% confidence bounds):
       p1 =        1.41  (-0.472, 3.292)
       p2 =      -3.821  (-9.636, 1.995)
```

We are visualising the above result using fit1, distance, and the specified loss to evaluate the fit's quality.

```
plot(fit2,nw,loss2)
```

With the creation of the new variable loss3 and the elimination of reliance,

```
loss3=loss2-fit2.p1*nw-fit2.p2
```

Number of walls

Wall count is one of the primary causes of propagation impairment. The intensity of the signal at the receiver end weakens with the number of walls present, and as the frequency rises, the loss through the walls also rises. As a result, the received power is only from less diffracted signal.

The number of walls is the second input parameter, and MATLAB's fit test was conducted for this parameter to get the optimum fit line.

```
fit3=fit(flr,loss3,'poly1')
```

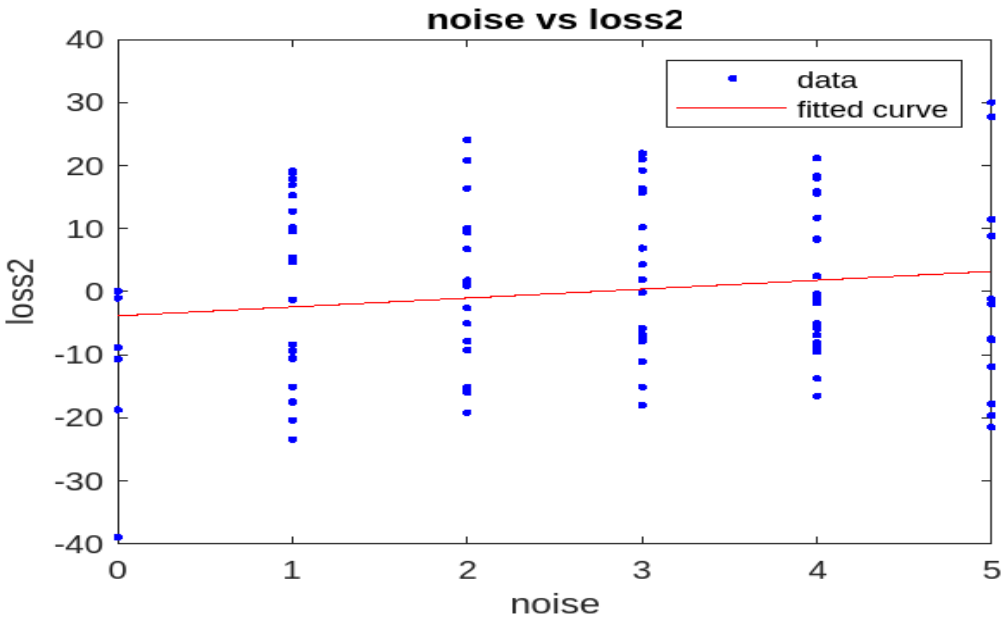


Figure 7

As a straight line is required in this situation, we are using "poly1," but we may also use "poly2" if a parabolic curve is required. After the MATLAB computations, the gradient and intercept are produced.

```
fit3 =

    Linear model Poly1:
    fit3(x) = p1*x + p2
    Coefficients (with 95% confidence bounds):
      p1 =       5.212  (4.434, 5.991)
      p2 =      -19.13  (-22.44, -15.82)
```

Now that dependence has been eliminated and a new variable, loss4, has been generated

```
loss4=loss3-fit3.p1*flr-fit3.p2
```

Estimated Loss

```
Estimated_Loss=fit1.p1*dist+fit2.p1*nw+fit3.p1*flr+(fit1.p2+fit
2.p2+fit3.p2)
```

The gradient and intercept are produced.

```
fit4 =

     Linear model Poly1:
     fit4(x) = p1*x + p2
     Coefficients (with 95% confidence bounds):
       p1 =      0.9617  (0.8666, 1.057)
       p2 =       3.195  (-4.905, 11.29)
```

plot(fit4,Estimated_Loss,loss)

Graph was plotted using finalfit due to its three-dimensional nature. A perfectly fitted curve may be shown by comparing the estimated loss to the provided loss, where there is less gap between the dispersed points and fitted curve. The dispersed points may now be shown to be closer to the fitted curve, demonstrating that a curve best approximates the trend on a scatter plot.

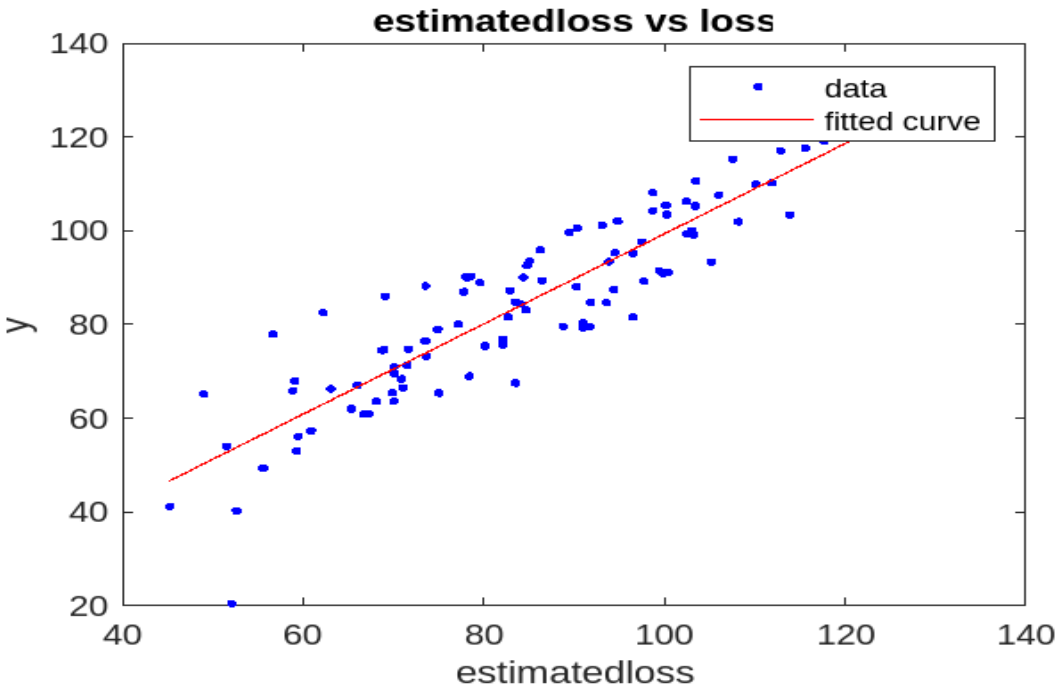


Figure 8

Root Mean Square Calculation:

Using the rms command in MATLAB, the rms values for loss, loss2, loss3, and loss4 were determined (lossn).

Following the elimination of recognised hazards, these numbers are computed to determine the residual loss.

R1=rms(loss)

=85.4598

R2=rms(loss2)

=14.098

R3=rms(loss3)

=13.9424

R4=rms(loss4)

=8.329

Coursework - 4

Task 1:

A) Calculate the scale height for oxygen pressure:

Given,
Air pressure ground= 1013pa
Height= 150hpa
h =12km
P(h)=Po exp (h/hs)
150=1013exp(-12/hs)
Taking ln,
Ln(150/1013)= In(exp(-12/hs)
-1.91= -12/hs
hs=12/1.91= 6.282 km.
Scale height, hs = 6.282 km.

B) Calculate the oxygen loss:

Oxygen loss =0.5680*6.282

= 3.568dB.

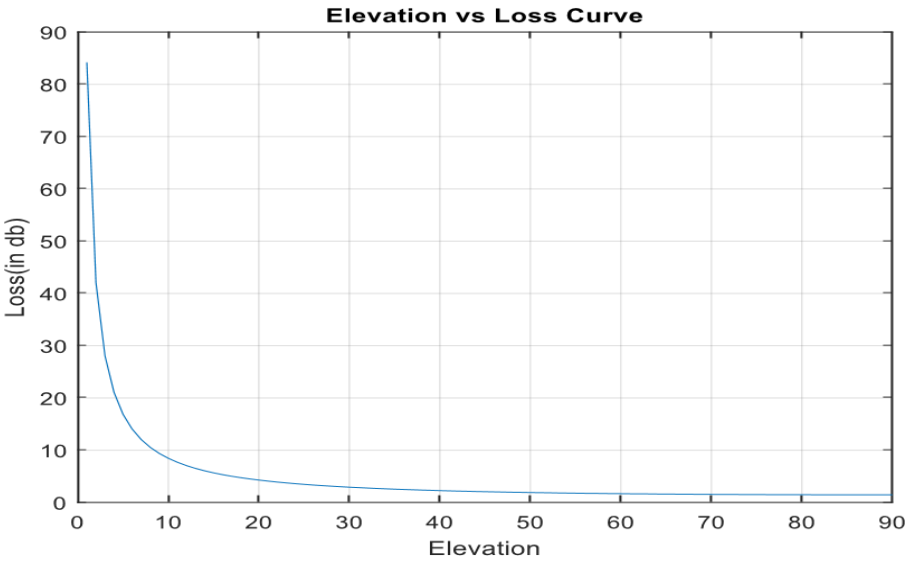


Figure 9

As observed in the preceding figure, a loss of 85 dB occurs at zero degrees elevation, whereas a loss of around 0.1 dB occurs at an angle of 900. The reason for this is that the loss was smaller since it was closer to the satellite at 900. As a result, the loss is reduced as the elevation angle increases.

D) Calculate the distance of Satellite:

distance = sat_dist(h,e)

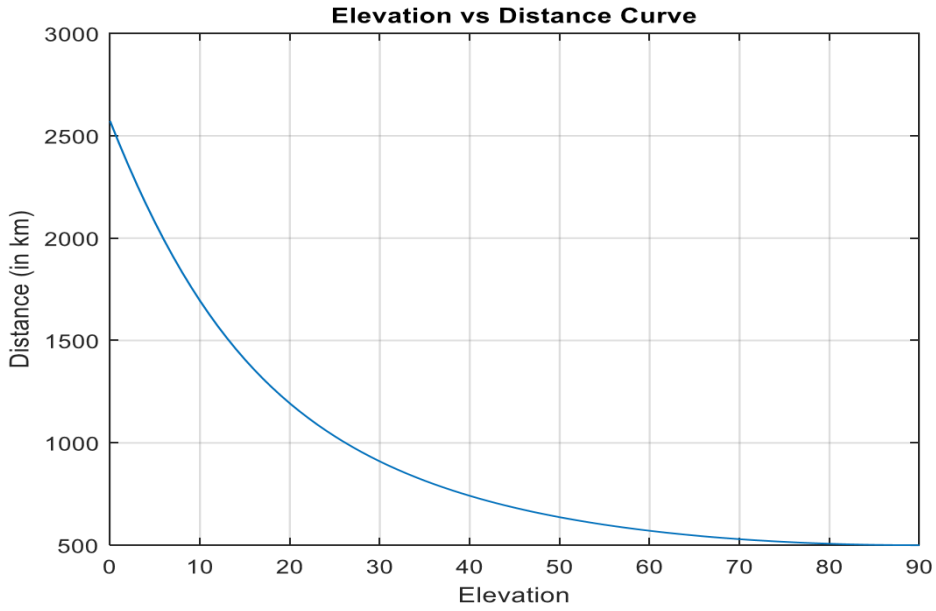
The distance between the observer and the satellite is calculated using the Matlab software

sat_dist.

Inputs,

Satellite orbit above the Earth h= 500km altitude in above Earth surface

e=Elevation in degree, (0 to90)



When the elevation angle is greater, the satellite is closer; if it is less, the satellite is farther away in the graph above.

E) Calculate the free-space loss and Total loss:

As stated in the question, the height is 500 kilometres, and the elevation angle is considered to range from 0 to 90. The loss of free space has been computed. As there are 91 angles between "0" and "91," the oxygen loss is computed using the linspace command, which also calculates the oxygen value (oxy_val) by dividing the oxygen loss by each elevation value. The overall loss is then determined by combining the oxygen value and free-space loss. Elevation data are taken in degrees by using the command "Sind" on the graph, which is drawn between the elevation and the overall loss.

```
distance = sat_dist(500,[0:1:90])
FSL=freespace(50000, distance)
oxy_loss= linspace(Oxygen_Loss,Oxygen_Loss,91)
Elevation= sind(0:1:90)
oxy_val= oxy_loss./Elevation
Total_loss= FSL+oxy_val
plot([0:1:90], Total_loss)
grid on
xlabel('Elevation')
ylabel('Total Loss (in dB)')
title('Elevation vs Total Loss Curve')
```

The overall loss reduces as the elevation angle increases in the graph below. This demonstrates that the loss from free space and oxygen will be smaller for higher elevation angles because the distance to the satellite will be shorter, and vice versa at lower elevation angles.

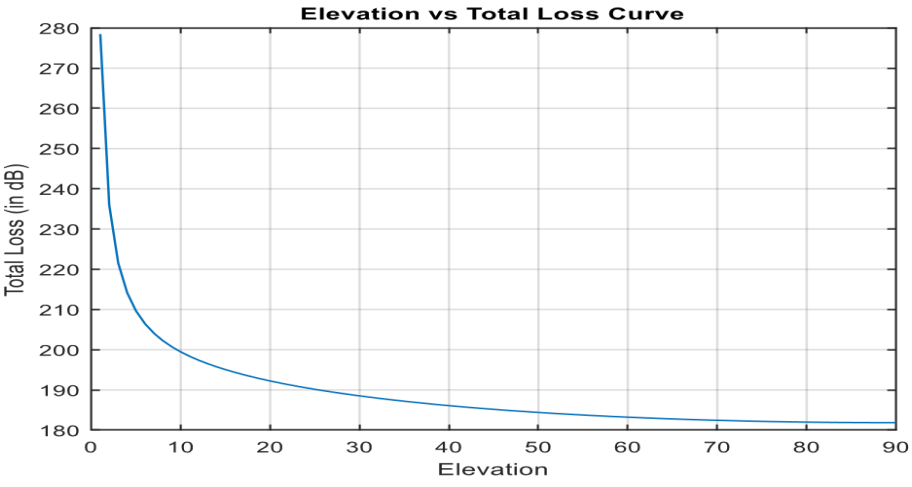


Figure 11

F) Calculate the minimum Elevation angle:

EIRP= PT+GR

$G_R = (4\pi A)/\lambda^2$

PR= PT + GT + GR – L ……Power Budget

$-130dB=[\ 20+10\log(4\pi*0.6361)/(0.006)^2]- L$

L = 203.4dB

Task-2:

SAMrain.m:

absorption = SAMrain(freqGHZ,elevation,rainrate,Hstation,Hice)

Hice = 7.8-0.1*Latitude

The ice height serves as the effective height of the rain while dealing with low rain rates, according to **Hice**

Input:

Freq(GHZ)=**35**

Elevation= **20^0**

Rainrate = The rainrate (mm/h) **27mm/h**

Hstation = height of ground station **0.062 km**

Hice = The magnitude of the latitude is determined by the height of the 0^0 isotherm (km), which is set to a negative integer.

Ap_leicester= 43.7452

Ap_bux = 40.2862

Analysis:

In this case, rain attenuation in Leicester is 43.75, which is higher than in Buxton, which holds 40.29. This is because there are less available signals in Leicester since the majority of them are attenuated by rain, which causes more route loss.

Task -3:

ITUrainattenuation.m:

Input:

Elevation = 64^0

Freq(GHZ)= 15

H0 = Above the sea level of earth station (km)

R001 = 0.01% of an average year's point rainfall rate for a place (mm/h). From ITU-R P.837 maps, this may be learned. percentphi: earth station's latitude (degrees)

Percentage = p=logspace(-1,-3,10)

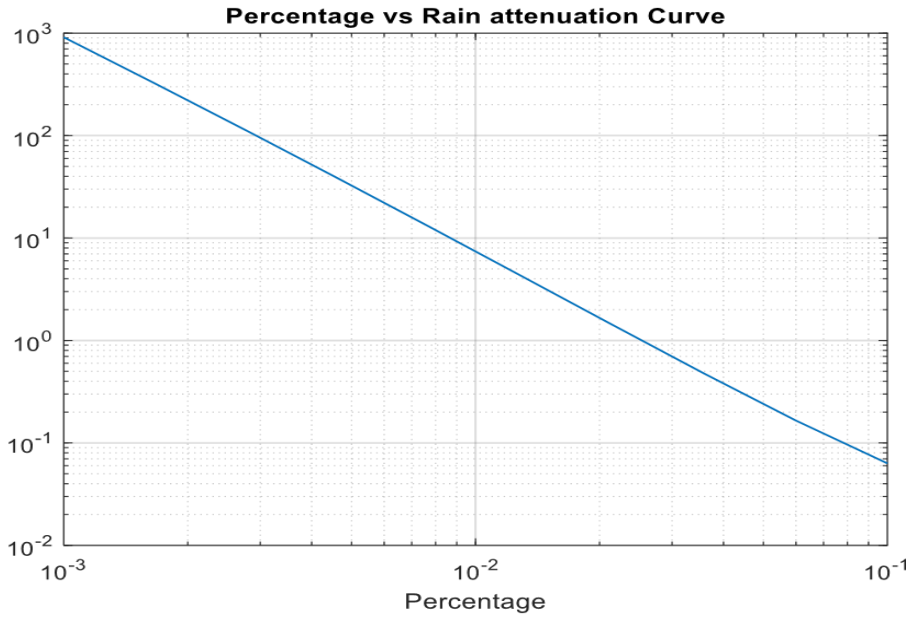


Figure 12

When the gain drops below 5 dB, no signal is obtained; the signal to noise ratio indicates the minimal signal that can be produced. It has been lowered by 7 dB. The proportion is around 0.01 at 7 dB. As can be seen in the graph above, rain attenuation increases as the percentage decreases. This is due to the fact that attenuation rises as elevation angle decreases. For instance, when the elevation angle is at 900, the satellite is closer, resulting in less attenuation; but, if the elevation angle is at 00, the satellite is now far away, resulting in greater attenuation.

B)What other effect might make this value inaccurate?

India's monsoon season is erratic, resulting in unequal rainfall across the nation and subsequent floods. This circumstance will cause the value to be incorrect.

Task -4:

ITUxpd.m

A) Cross polar Discrimination:

Freq(GHZ)= 15GHZ

Tau= tilt angle of the horizontal electric field vector when it is linearly polarized

Theta=elevation angle 20deg

Ap= over the requisite % of time, rain attenuation (dB),**p=25-20=5dB**

P= 0.01%

XPDp= ITUxpd(15,20,30,5,0.01)

= 20.821

Cross Polar discrimination, **XPD= 20.82**

B) Finding SNR:

SNR= 20dB
the dB to watts conversion
SNR = 100W
S/NR= S/N N/S/SNR N/S/100
S/I = 100 I = S/100, or signal interference
SINR = S/(N+I) N+I = (S/100) + (S/100) = 2S/(S/100)
SINR is equal to S / (S / 100)
SINR = 50S/2S SINR = 100S.
SINR is equal to **16.98 dB** in decibels.